

Building a Problem Space for Science Translation Advisory Group and Project Overview

Purpose

This Army Research Office-funded project examines how scientific findings move from research settings into real-world use. The project will build a structured problem space for science translation by identifying the actors, contexts, barriers, facilitators, and leverage points that shape whether research becomes usable policy, programs, technologies, practices, or capabilities

The project is designed to produce a structured representation of the science translation ecosystem. Expected products include:

- **A science translation problem map:** A structured map of the major barriers, facilitators, stakeholder roles, contextual factors, and coordination needs that shape research translation.
- **A stakeholder-informed framework:** A framework showing how translation processes vary across defense, academia, industry, and adjacent communities, and where coordination breaks down or can be improved.
- **Priority areas and leverage points:** Identification of the problems, relationships, and intervention points stakeholders view as most important, feasible, and actionable
- **Briefings, reports, and scholarly products:** Project outputs intended to support future research, program planning, stakeholder coordination, and practical improvements in how research findings are moved toward use.

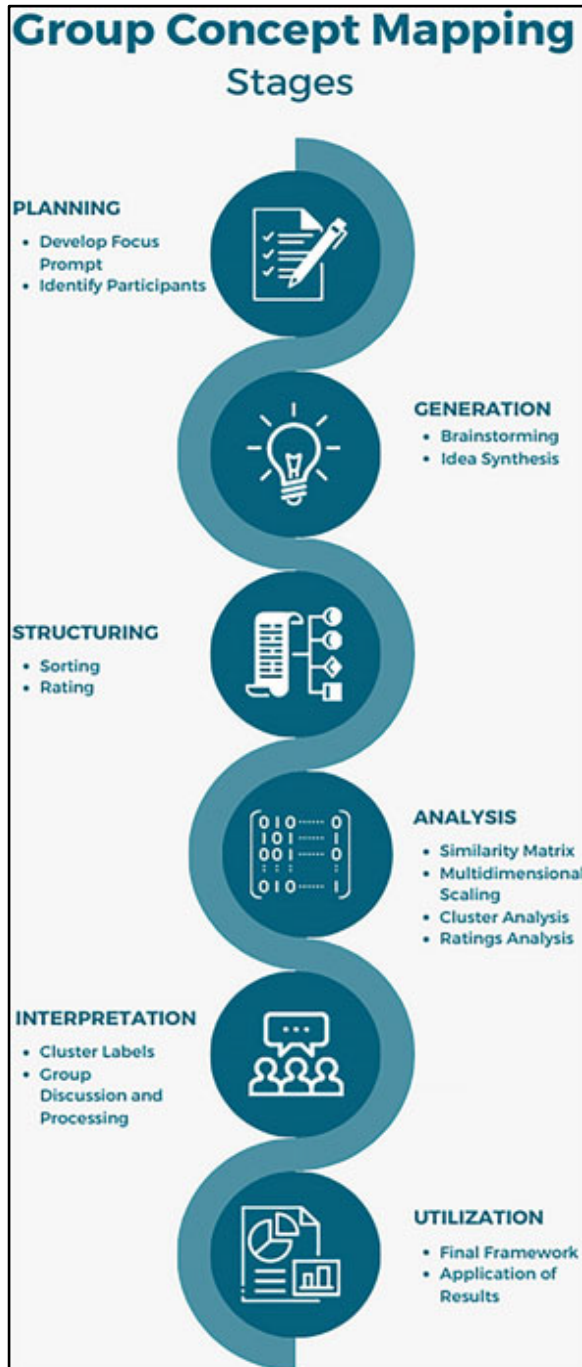
How the project will work

The project uses Group Concept Mapping, a structured method for eliciting, organizing, rating, and interpreting stakeholder perspectives. The figure below outlines the stages and activities of the GCM process. In general, participants generate statements about a complex problem, sort those statements into meaningful groups, rate them on key dimensions, and help interpret the resulting maps and themes.

The current project is structured around three primary stakeholder pillars: defense, industry, and academia. Participant engagement will occur through three groups: an advisory group, a core group, and an extended group. The advisory group will contribute to refining the project framing and stakeholder engagement strategy. The core group will support in-depth discussion and sensemaking, providing input through brainstorming, sorting, and rating activities. The extended group will provide broader asynchronous input through virtual brainstorming, sorting, rating, and feedback activities.

Role of the advisory group

The advisory group is a small group of trusted thought leaders who help ensure that the project asks the right questions, engages the right communities, and produces useful outputs. Advisors are being asked to be objective and to help sharpen the project.



Advisory input will help the project team:

- ❖ Refine the translation problem framing and focus prompt
- ❖ Identify stakeholder communities that need to be represented
- ❖ Advise on recruitment strategy for core and extended stakeholder groups
- ❖ Review emerging maps, themes, or summaries for credibility and usefulness
- ❖ Articulate practical implications for research programs, policy, coordination, and implementation

Advisory Recognition

The central risk in studying science translation is producing a framework that is internally coherent but disconnected from how translation actually happens. Advisory members help prevent that by bringing high-level experience with research systems, programmatic decision-making, operational needs, implementation pathways, and institutional constraints.

Advisory contributions will be recognized in ways appropriate to the level and type of contribution. Publication authorship, if applicable, will follow standard scholarly norms and will depend on substantive intellectual contributions to the work.

Figure 1. From Napier-Raman, S., Rosas, S., Hossain, S. Z., Mpofu, E., Lee, M., Liamputtong, P., Dune, T., Mapedzahama, V. (2023). Concept Mapping Method. In P. Liamputtong (Ed.). Handbook of Social Sciences and Global Public Health (pp. 1-20). Springer, Switzerland.

Project Personnel:

Stephen M. Fiore - University of Central Florida - sfiore@ucf.edu

Rhyse Bendell - University of Central Florida - rhyse.bendell@ucf.edu

Scott Rosas - Concept Systems Inc. - srosas@conceptsistemas.com

Greg Ruark - Army Research Office - gregory.a.ruark.civ@army.mil